

Self-Fulfilling Debt Crises with Long Stagnations

Ayres, Navarro, Nicolini & Teles (JPE 2025)

Kacper Makowski

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Roadmap

- ① Motivation
- ② Where it sits in the literature
- ③ The Calvo mechanism
- ④ Model
- ⑤ Why the bimodal process matters
- ⑥ Quantitative results
- ⑦ Policy & takeaways

Two crises

- Argentina 2001 — spreads 93-99 $\sim 7\%$, default, collapse.
- Southern Europe 2010–12 — spreads $> 5\%$, *no* default.
- OMT / “whatever it takes” (2012) — spreads fall, no intervention, fundamentals unchanged.

The question

Fundamentals vs. Self-fulfilling expectations

- Both fit a *single* crisis.
- Quantitative question: *how much?* *when?*

Three families

- ① **Fundamentals** — Eaton–Gersovitz, Arellano. Unique eq.
- ② **Rollover runs** — Cole–Kehoe; Bocola–Dovis; Aguiar et al. Sudden default.
- ③ **Interest-rate multiplicity** — Calvo (1988). Price & default jointly set.

This paper: Calvo source + Eaton–Gersovitz engine.

Contribution

Quantitative default model + Calvo multiplicity + bimodal growth
⇒ robust crises, only in stagnations.

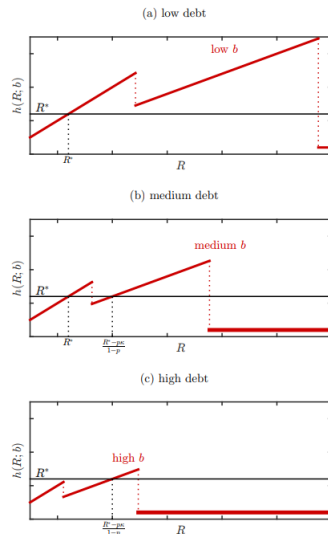
Unimodal income ⇒ high-rate schedule slopes down ⇒ fragile / ruled out.

Calvo intuition

$$R^* = h(R; b) = [1 - \text{Pr}(\text{default})] R + \text{Pr}(\text{default}) \kappa$$

- Expect repayment $\Rightarrow$ low R
- Low $R \Rightarrow$ light debt burden
- Light burden $\Rightarrow$ repayment
- Expect default $\Rightarrow$ high R
- High $R \Rightarrow$ heavy debt burden
- Heavy burden $\Rightarrow$ default

Figure 1: Expected return function for different levels of debt

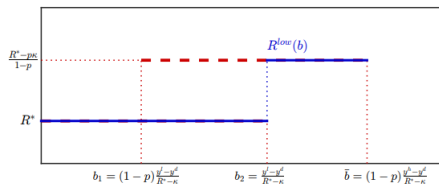


The interest-rate correspondence

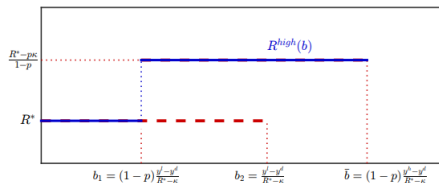
- $b < b_1$: only the low (risk-free) rate.
- $b_1 < b < b_2$: **two** rates — low *or* high (multiplicity).
- $b_2 < b < \bar{b}$: only the high rate. $b > \bar{b}$: no lending.

Figure 2: Interest rate schedules

(a) low interest rate schedule



(b) high interest rate schedule



Environment

- SOE, output **grows**: $Y_t = \Gamma_t e^{\sigma \epsilon_t}$, $\Gamma_t = g_t \Gamma_{t-1}$.
- $g_t \in \{g_H, g_L\}$ two-state Markov — **growth vs. stagnation**.
- Long-term debt (decay δ); **borrower chooses current issuance N** .
- No commitment; default $\Rightarrow$ exclusion + output loss $1 - \phi(g)$.

Pricing $\Rightarrow$ multiplicity

$$1 = \frac{R(N, b, g, s)}{1 + r^*} \mathbb{E}[Q(b', g', s', \epsilon') \mid g, s], \quad gb' = (1 - \delta)b + R N$$

- High $R \Rightarrow$ high $b' \Rightarrow$ high default prob $\Rightarrow$ low $\mathbb{E}Q \Rightarrow$ justifies high R .
- Sunspot s selects high vs. low schedule where both exist.
- Government chooses issuance (N), lenders react with R .

Equilibrium selection: the sunspot

- $s \in \{s_G, s_B\}$ selects good (low-rate) or bad (high-rate) schedule.
- i.i.d.; p_B = prob. of the bad sunspot = how pessimistic lenders are.
- p_B is *not* estimated — it is set (it is, by definition, arbitrary).

Fragility under unimodal distributions

- Commonly-used (unimodal) income: high-rate schedule **slopes down** in b .
- Then $R(b)$ falls as b rises $\Rightarrow$ borrower never sits there $\Rightarrow$ ruled out.
- **Bimodal** (good & bad times, low σ) $\Rightarrow$ **upward** high-rate schedule, robust.

The bimodal process is in the data

- Estimate Markov-switching growth (Kim 1994 filter): AR, BR, IT, PT, ES.
- All bimodal: $g_H - g_L \approx 6$ pp on average ($> 3 \times$ transitory σ).
- Both regimes **persistent**: 60–80%. Stagnations are long.

Table 1: Prior and posterior distributions

	$\ln(g_L)$	$\ln(g_H)$	p_L	p_H	σ
Prior distribution					
	$U[-0.1, 0.1]$	$U[-0.1, 0.1]$	$U[0.1, 1.0]$	$U[0.1, 1.0]$	$U[10^{-3}, 0.5]$
Posterior distribution (mean, and 5th to 95th percentile intervals)					
Countries					
Italy	-0.017 [-0.037,-0.008]	0.022 [0.018,0.028]	0.646 [0.050,0.990]	0.843 [0.627,0.990]	0.016 [0.012,0.023]
Portugal	-0.002 [-0.011,0.003]	0.048 [0.041,0.057]	0.805 [0.516,0.990]	0.720 [0.454,0.939]	0.019 [0.014,0.025]
Spain	-0.018 [-0.026,-0.010]	0.033 [0.026,0.039]	0.629 [0.308,0.990]	0.838 [0.653,0.990]	0.017 [0.013,0.025]
Argentina	-0.040 [-0.049,-0.022]	0.060 [0.051,0.078]	0.620 [0.346,0.877]	0.581 [0.358,0.781]	0.033 [0.025,0.044]
Brazil	-0.033 [-0.071,-0.022]	0.029 [0.025,0.032]	0.589 [0.103,0.860]	0.793 [0.627,0.923]	0.019 [0.014,0.025]

Calibration

- Two calibrations: **Argentina** ($p_B = 25\%$) and **Spain** ($p_B = 1\%$).
- $\beta = 0.75$, $\gamma = 3$, $r^* = 3.5\%$, $\kappa = 0.75$; maturity, $\phi(g)$ country-specific.
- p_B set first; default costs then matched to debt & pre-crisis spreads.

Table 2: Benchmark calibration

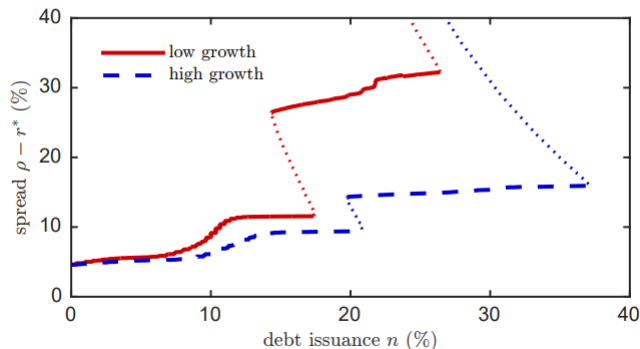
A. Parameters that are common across regions			
Description	Parameter	Value	
discount factor	β	0.75	
risk aversion	γ	3.0	
risk-free rate	R^*	1.035	
re-entry probability	$1 - \theta$	0.10	
fraction of debt recovered after default	κ	0.750	
B. Parameters that vary across regions			
Description	Parameter	Argentina	Spain
inverse of average maturity	δ	0.40	0.15
probability of remaining in low growth	p_L	0.600	0.629
probability of remaining in high growth	p_H	0.750	0.838
low-growth rate	$\ln(g_L)$	-0.033	-0.018
high-growth rate	$\ln(g_H)$	0.029	0.033
standard deviation of transitory shock	σ	0.019	0.017
default cost in high growth state	$\phi(g_H)$	0.90	0.945
default cost in low growth state	$\phi(g_L)$	0.99	0.935
probability of bad sunspot	$p_B \equiv 1 - p_C$	0.25	0.01

State dependence: the central result

multiplicity region **large** in low growth, **tiny** in high growth

- Low growth: stagnation likely to persist $\Rightarrow$ pessimism self-validates.
- High growth: switch to stagnation unlikely $\Rightarrow$ no robust bad eq.

Figure 6: Interest rate spreads for Argentina ($b = 20\%$)



Expectations matter: vary p_B

- Argentina p_B : $25\% \rightarrow 1\% \Rightarrow$ default rate $5.1\% \rightarrow 0.5\%$, spread $16.4\% \rightarrow 0.2\%$.
- Spain p_B : $1\% \rightarrow 25\% \Rightarrow$ spread $1.2\% \rightarrow 7.7\%$ (low-growth $0.1\% \rightarrow 9.7\%$).
- Same fundamentals — beliefs alone flip the outcome.

Expectations matter: vary p_B - tables

Table 3: Simulation moments: Argentina

	Benchmark ($p_B = 25\%$)	$p_B =$
First moments (%)		
avg(b/y)	30	2
avg($qb/\delta y$)	49	6
avg(f/y)	50	6
avg(n/y)	25	2
avg(tb/y)	5.1	1
default rate	5.1	0
avg($spread$)	16.4	0
Low-growth state		
avg(b/y)	22	2
avg($qb/\delta y$)	11	6
avg(f/y)	37	6
avg(n/y)	25	2
avg(tb/y)	-3.0	4
default rate	13.3	1
avg($spread$)	30.4	0
High-growth state		
avg(b/y)	30	2
avg($qb/\delta y$)	50	6
avg(f/y)	50	5
avg(n/y)	25	2
avg(tb/y)	5.1	0

Table 4: Simulation moments conditional on not being in default: Spain

	Benchmark ($p_B = 1\%$)	$p_B = 25\%$
First moments (%)		
avg(b/y)	17	14
avg($qb/\delta y$)	88	53
avg(f/y)	86	52
avg(n/y)	18	12
avg(tb/y)	-1.4	1.6
default rate	5.2	6.1
avg($spread$)	1.2	7.7
Low-growth state		
avg(b/y)	16	12
avg($qb/\delta y$)	80	35
avg(f/y)	84	47
avg(n/y)	17	12
avg(tb/y)	-0.9	0.5
default rate	18	20
avg($spread$)	0.1	9.7
High-growth state		
avg(b/y)	17	14
avg($qb/\delta y$)	89	55
avg(f/y)	86	53
avg(n/y)	18	12
avg(tb/y)	-1.5	1.6

Endogenous austerity & gambling for redemption

- **Austerity:** issuance n falls as debt service b rises — dodge the rate jump.
- **Gambling:** borrow *past* the jump for current consumption (Argentina).
- **Spain:** > 99% of defaults on high→low transitions; austerity keeps high spreads off-equilibrium.

Figure 10: Policy functions and equilibrium spreads for Argentina

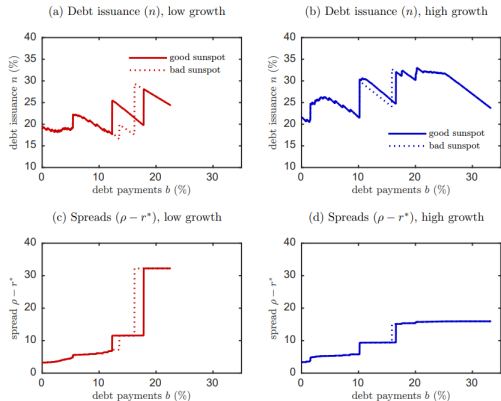
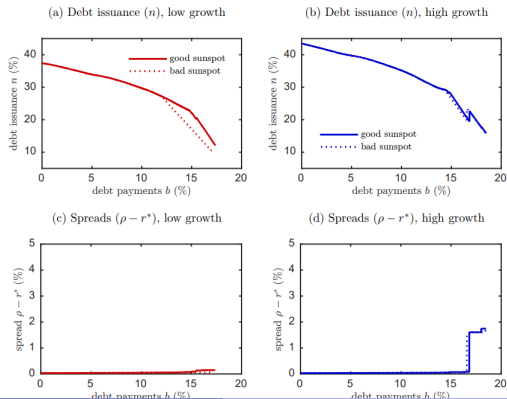


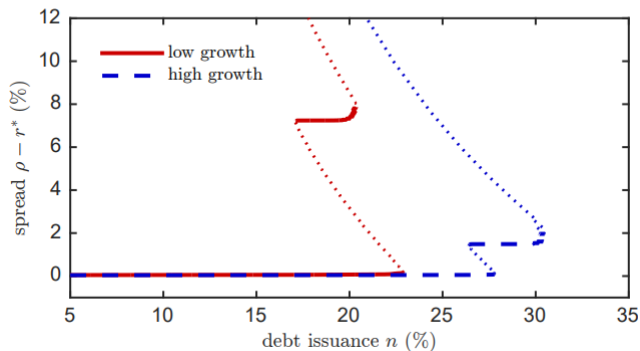
Figure 11: Policy functions and equilibrium spreads for Spain



Spain: schedules and the 2012 gap

- Low-growth multiplicity for issuance $\approx 17\text{--}22\%$ of GDP.
- High–low spread gap in that region $\approx 7\%$ (*not calibrated*).
- Close to Spain's observed peak spread ($\gtrsim 6\%$, July 2012).

Figure 8: Interest rate correspondence for Southern Europe ($b = 15\%$)



Policy

- Backstop at the **good-schedule** rate $\Rightarrow$ removes the bad equilibrium.
- Works in **stagnations**, where multiplicity lives — OMT logic.
- Caveat: useless where the high spread is **fundamental**.

Takeaways

- ① Calvo multiplicity + Eaton–Gersovitz engine; borrower picks *current* issuance.
- ② Robust multiplicity needs a **bimodal, persistent** growth process.
- ③ Crises are **state-dependent**: stagnations only.
- ④ Expectations (p_B) are **first-order** for Argentina vs. Spain.

Key references

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- Eaton & Gersovitz (1981); Arellano (2008); Aguiar & Gopinath (2006).